

A cost-effectiveness analysis of HCM sudden cardiac death risk algorithms for ICD

Priment Statistics and Methodology seminar

Nathan Green

Outline

- Background
- Problems
- Results
- Current & Future Work
- Conclusion

Resources

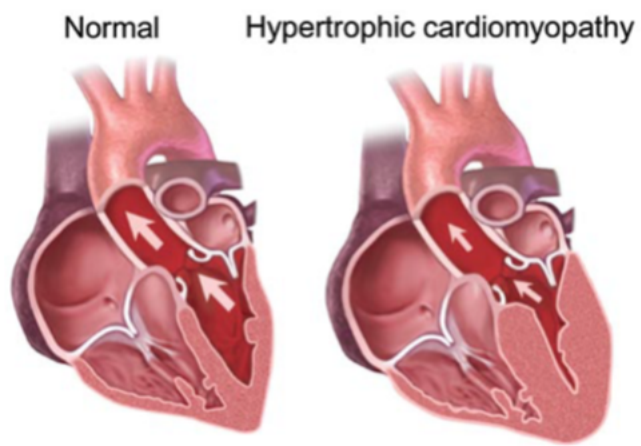
Slides and code here: github.com/n8thangreen/HCM-ICD-cost-eff-model-talk

Remember to introduce myself and the topic here. emerging approaches at the interface of data science, statistics, and public health. how data-driven methods are shaping real-world health impact.

Background

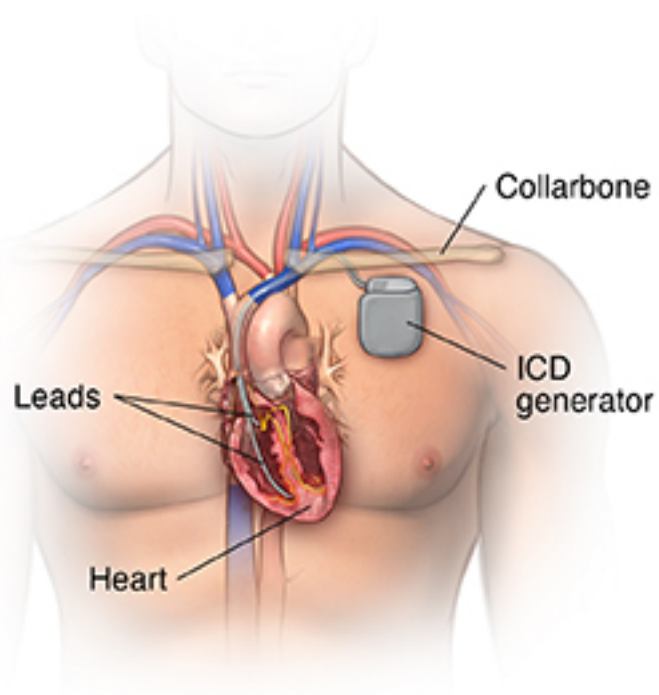
Hypertrophic cardiomyopathy (HCM)

HCM is the most common heritable cardiac condition and is the most common cause of sudden cardiac arrest in the young, affecting about 1 in 500 people. Familial hypertrophic cardiomyopathy is a heart condition characterized by thickening (hypertrophy) of the heart muscle, more specifically the ventricle. The thickened heart muscle, can make it challenging to keep up with the oxygenation demands of the body and the heart muscle itself. Many people with HCM have few, if any, symptoms and can lead normal lives without significant symptoms. However, this condition can also have serious consequences. Life threatening arrhythmias resulting in cardiac arrest can sometimes be the first symptom.



Symptoms include: Shortness of breath, Chest pain, Fainting, Palpitations

Implantable Cardioverter-Defibrillator (ICD)



An ICD is a small battery-powered device placed under the skin—usually near the collarbone. It constantly monitors your heart rhythm and delivers life-saving electrical shocks or pacing to correct dangerously fast or irregular heartbeats (arrhythmias), preventing sudden cardiac arrest

Previous work



European Heart Journal (2014) 35, 2010–2020
doi:10.1093/eurheartj/ehu439

FASTTRACK CLINICAL RESEARCH
Myocardial disease

A novel clinical risk prediction model for sudden cardiac death in hypertrophic cardiomyopathy (HCM Risk-SCD)

Constantinos O'Mahony¹, Fatima Jichi², Menelaos Pavlou⁸, Lorenzo Monserrat³, Aristides Anastasakis⁴, Claudio Rapezzi⁵, Elena Biagini⁵, Juan Ramon Gimeno⁶, Giuseppe Limongelli⁷, William J. McKenna¹, Rumana Z. Omar^{2,8} and Perry M. Elliott^{1*}, for the Hypertrophic Cardiomyopathy Outcomes Investigators

O'Mahony et al. (2014)

Problem

- Hypertrophic cardiomyopathy (HCM) is a leading cause of sudden cardiac death (SCD) in young adults. Current risk algorithms provide only a crude estimate of risk and fail to account for the different effect size of individual risk factors.
- The aim of this study was to develop and validate a new SCD risk prediction model that provides individualized risk estimates.

Data

	All	The Heart Hospital, UK	A Coruña University Hospital, Spain	1st Department of Cardiology, University of Athens, Greece	Institute of Cardiology, Bologna, Italy	University Hospital Virgen de la Arrixaca, Spain	Monaldi Hospital, Italy
Baseline							
Number of patients	3675*	1593 (43%)	590 (16%)	474 (13%)	456 (12%)	406 (11%)	156 (4%)
Male	2349 (64%)	1,018 (64%)	364 (62%)	340 (72%)	292 (64%)	243 (60%)	92 (59%)
Age, years ^b	48 ± 17	43 ± 15	57 ± 15	47 ± 16	50 ± 17	53 ± 17	44 ± 16
NYHA III/IV	426 (12%)	136 (8.5%)	76 (13%)	77 (16%)	41 (9%)	82 (20%)	14 (9%)
Myectomy	34 (1%)	17 (1%)	6 (1%)	1 (0.2%)	5 (1%)	2 (0.5%)	3 (2%)
Alcohol septal ablation	10 (0.3%)	4 (0.3%)	2 (0.3%)	0	2 (0.4%)	2 (0.5%)	0
Amiodarone	468 (13%)	217 (14%)	79 (13%)	44 (9%)	78 (17%)	39 (10%)	11 (7%)
ICD	42 (1%)	14 (1%)	6 (1%)	1 (0.2%)	5 (1%)	8 (2%)	8 (5%)
Permanent /persistent AF	366 (10%)	98 (6%)	133 (23%)	39 (8%)	19 (4%)	65 (16%)	12 (8%)
NSVT	634 (17%)	300 (19%)	82 (14%)	56 (12%)	80 (18%)	85 (21%)	31 (20%)
LA diameter, mm ^b	44 ± 8	44 ± 8	45 ± 8	44 ± 6	46 ± 9	44 ± 8	45 ± 8
LVOTG _{max} , mmHg ^b	12 (5–49)	9 (5–46)	10 (7–45)	10 (4–60)	20 (15–50)	4 (4–60)	4 (4–40)
LVedd, mm	45 ± 7	44 ± 6	45 ± 6	47 ± 5	44 ± 7	44 ± 7	46 ± 7
MWT, mm ^b	20 ± 5	20 ± 6	20 ± 5	18 ± 4	20 ± 5	20 ± 5	20 ± 5
FS, % ^{b,c}	41 ± 9	42 ± 9	41 ± 9	41 ± 7	41 ± 11	39 ± 10	32 ± 11
FHSCD	886 (24%)	482 (30%)	42 (7%)	147 (31%)	69 (15%)	110 (27%)	36 (23%)
Unexplained syncope	507 (14%)	274 (17%)	55 (9%)	90 (19%)	11 (2%)	55 (14%)	22 (14%)
Follow-up							
Enrolment period	1972–2011	1988–2005	1983–2010	1993–2011	1972–2011	1983–2009	2001–2010
End of follow-up period	2012	2010	2011	2012	2011	2011	2011
Total patient-years	24 313.4	11 779.4	4009.3	3709.8	2955.7	1274.2	584.6
Median follow-up, years	5.7 (2.8, 9.2)	6.6 (4.2, 10.0)	6.0 (3.5, 9.2)	7.4 (4.5, 11.1)	3.4 (1.2, 8.8)	2.5 (0.8, 5.4)	3.8 (1.6, 5.4)
SCD endpoints	198 (5%)	89 (6%)	20 (3%)	25 (5%)	41 (9%)	20 (5%)	3 (2%)
5-year cumulative hazard; % (95% CI)	3.8% (3.1, 4.5)	3.8% (2.9, 5.0)	1.6% (0.8, 3.3)	3.4% (2.1, 5.6)	6.1% (3.8, 9.7)	5.5% (3.3, 9.0)	2.0% (0.7, 6.2)

Results

- The prognostic model was derived from a retrospective, multi-centre longitudinal cohort study. The model was developed from the entire data set using the Cox proportional hazards model and internally validated using bootstrapping.
- The cohort consisted of 3675 patients from six centres. During a follow-up period of 24,313 patient-years (median 5.7 years), 198 patients (5%) died suddenly or had an appropriate ICD shock.
- Of eight pre-specified predictors, associated with SCD/appropriate ICD shock at the 15% significance level age, maximal left ventricular wall thickness, left atrial diameter, left ventricular outflow tract gradient, family history of SCD, non-sustained ventricular tachycardia, and unexplained syncope
- final model to estimate individual probabilities of SCD at 5 years.
 - For every 16 ICDs implanted in patients with 4% 5-year SCD risk, potentially 1 patient will be saved from SCD at 5 years.

Standard Failure Probability

Define the Prognostic Index (PI) using the coefficients from the standard Cox model

$$PI = 0.159 \times \text{Maximal wall thickness} \quad (1)$$

$$- 0.003 \times \text{Maximal wall thickness}^2 \quad (2)$$

$$+ 0.026 \times \text{Left atrial diameter} \quad (3)$$

$$+ 0.004 \times \text{Maximal LVOT gradient} \quad (4)$$

$$+ 0.458 \times \text{Family history SCD} \quad (5)$$

$$+ 0.826 \times \text{NSVT} \quad (6)$$

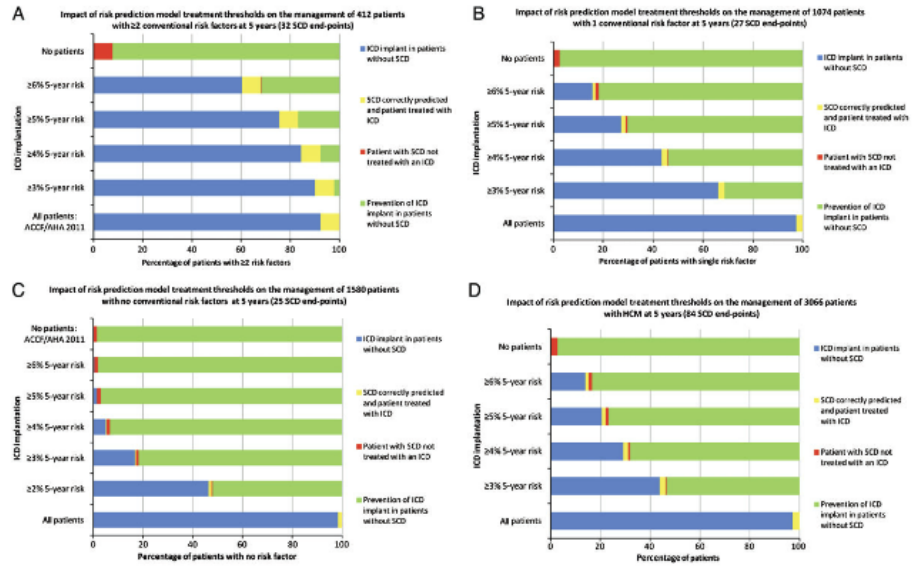
$$+ 0.717 \times \text{Unexplained syncope} \quad (7)$$

$$- 0.018 \times \text{Age} \quad (8)$$

Risk Calculator

$$\hat{P}_{\text{SCD at 5 years}} = 1 - 0.998^{\text{exp}(PI)}$$

Impact on Decision-Making



A cost-effectiveness analysis of hypertrophic cardiomyopathy sudden cardiac death risk algorithms for implantable cardioverter defibrillator decision-making

Nathan Green ^{1,*}, Yang Chen ², Constantinos O'Mahony^{3,4}, Perry M. Elliott^{3,4}, Roberto Barriales-Villa ⁵, Lorenzo Monserrat ⁵, Aristides Anastasakis⁶, Elena Biagini ⁷, Juan Ramon Gimeno⁸, Giuseppe Limongelli ⁹, Menelaos Pavlou¹⁰ and Rumana Z. Omar¹⁰

Green et al. (2024)

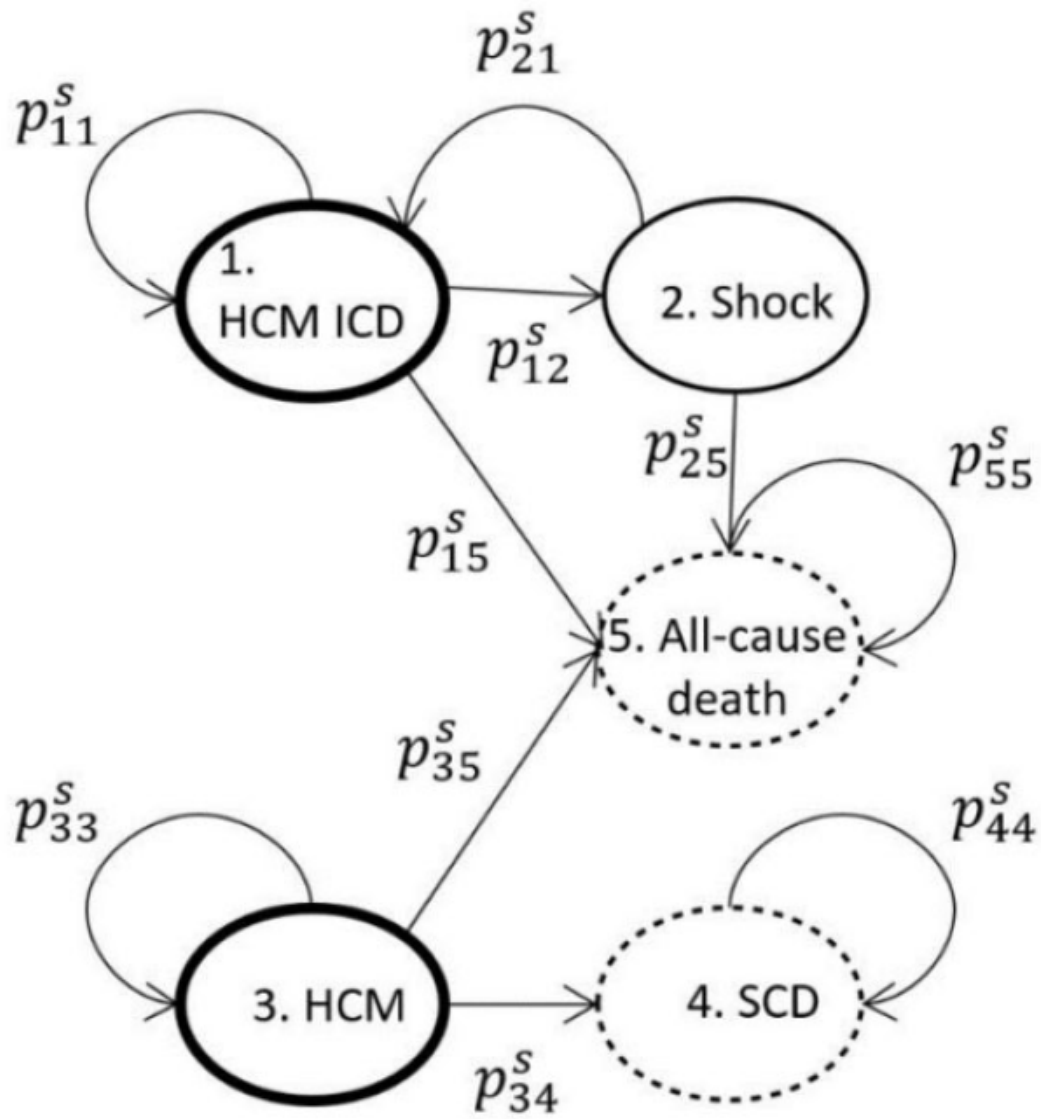
Aims

- To conduct a contemporary cost-effectiveness analysis examining the use of implantable cardioverter defibrillators (ICDs) for primary prevention in patients with hypertrophic cardiomyopathy (HCM)

Methods

- A discrete-time Markov model was used to determine the cost-effectiveness of different ICD decision-making rules for implantation.
- Several scenarios were investigated, including the reference scenario of implantation rates according to observed real-world practice.
- A 12-year time horizon with an annual cycle length was used.
- Transition probabilities used in the model were obtained using Bayesian analysis.

Markov Model



Input Data

Description	Parameter	Value	Source
<i>Health</i>			
HCM without ICD	q_hcm	0.88 QALY/year	Sanders et al. [2005]
Manage with ICD proportion	u_icd	0.9	Magnusson and Wimo [2020]; Holbrook et al. [2020]
Death	q_death	0 QALY/year	
Implantation procedure decrement	u_implant	−0.048	Holbrook et al. [2020]
Implantation complication decrement	u_compl	−0.096	Holbrook et al. [2020]
Shock utility proportion	u_shock	0.875	Buxton et al. [2006]
<i>Cost</i>			
ICD appointment	c_appt	£145	(WF02A) [NHS England, 2021]
Perform risk score	c_rs	£0	
Implant ICD	c_icd	£4666	(EY02B) [NHS England, 2021]
Implant complication	c_compl	£28 857	Formula derived
Non-fatal shock with hospitalisation	c_shock	£165	Thijssen et al. [2014]
Lead infection	c_inf	£37 116	Thijssen et al. [2014]
Lead dislodgement	c_dis	£6146	Thijssen et al. [2014]
HCM without ICD	c_hcm	0	
Sudden cardiac death (SCD)	c_scd	0	
All-cause death	c_death	0	
<i>Probabilities</i>			
Initial implant complication	p_compl	0.043	Cunningham et al. [2012]
Lead infection	p_inf_init	0.02 277	Thijssen et al. [2014]
Lead dislodgement	p_dis_init	0.00 828	Thijssen et al. [2014]
Cohort size	N	3672	
Time horizon	T	12 years	Expert input
Annual number of appointments	n_appt	2	Expert input

Transition Probability Matrix

$$\begin{pmatrix} p_{11}^s & p_{12}^s & 0 & 0 & p_{15}^s \\ 1 - p_{15}^s & 0 & 0 & 0 & p_{15}^s \\ 0 & 0 & p_{33}^s & p_{34}^s & p_{35}^s \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Model Equations

Denote x as the observed number of transitions, p the probability of a transition and n as the total number of transitions from a given state. The hyperparameters α characterise the prior knowledge on p . Superscripts indicate the decision rule used.

i Likelihood

$$x_{i.}^{(1)} \sim \text{Multinomial}(p_{i.}^{(1)}, n_i^{(1)}), \quad i = 1, 3 \quad (9)$$

$$x_{i.}^{(2)} \sim \text{Multinomial}(p_{i.}^{(2)}, n_i^{(2)}), \quad i = 1, 3 \quad (10)$$

$$(11)$$

i Priors

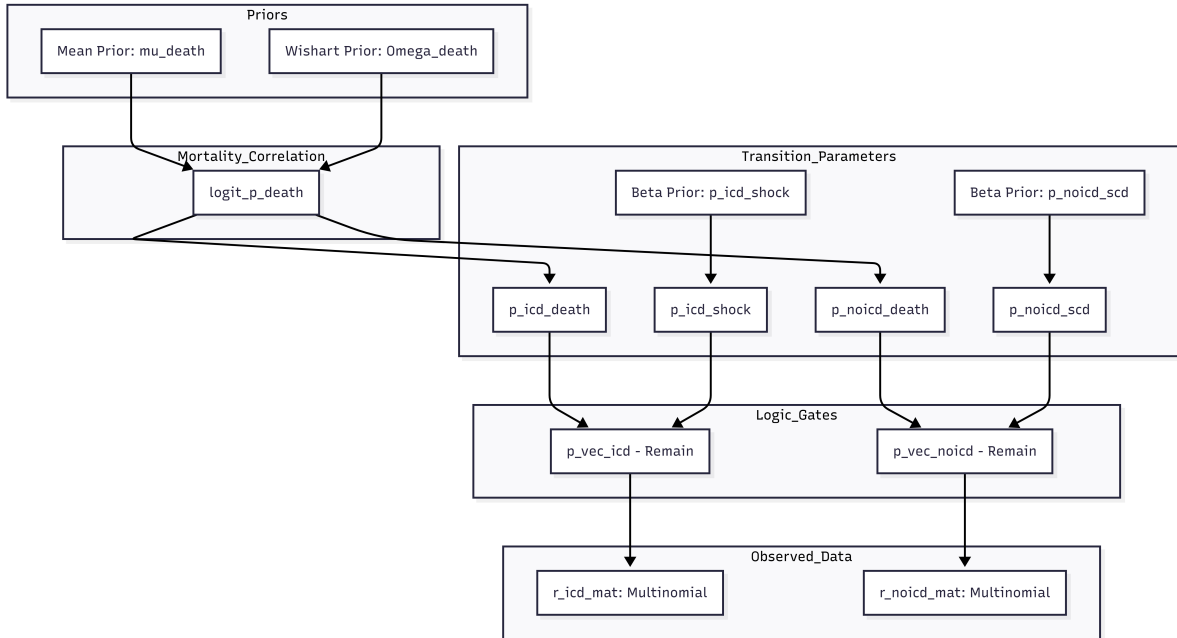
$$p_{i.}^{(1)} \sim \text{Dirichlet}(\alpha^{(1)}), \quad i = 1, 3 \quad (12)$$

$$p_{i.}^{(2)} \sim \text{Dirichlet}(\alpha^{(2)}), \quad i = 1, 3 \quad (13)$$

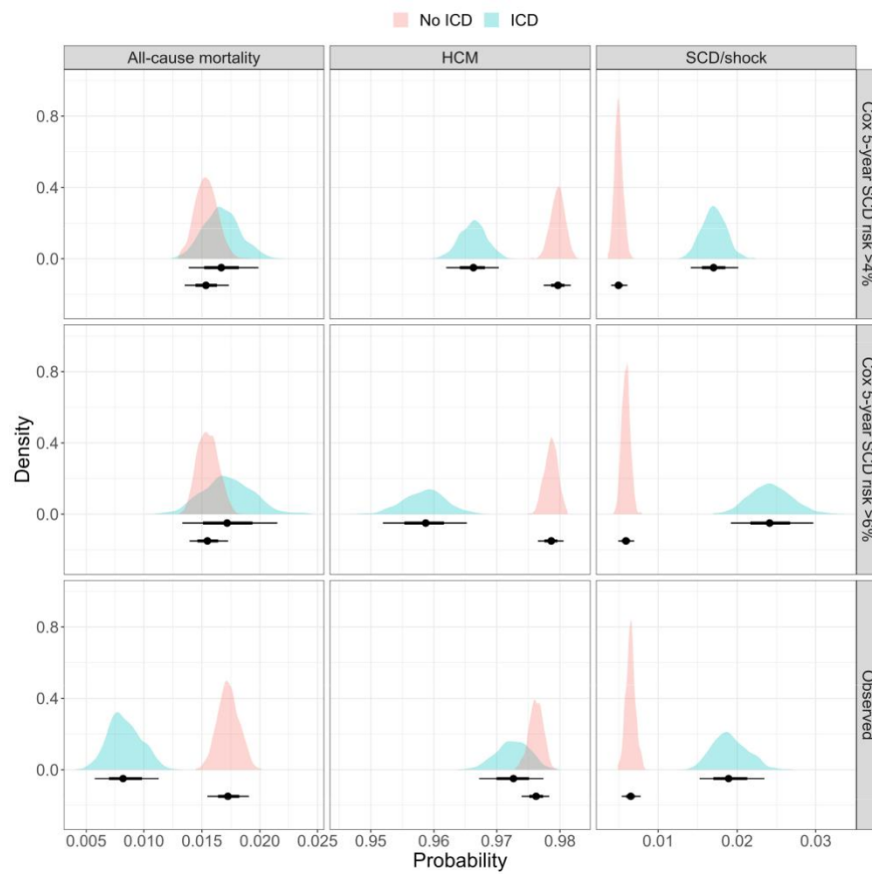
For all final sink states,

$$p_{ij}^{(s)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

DAG



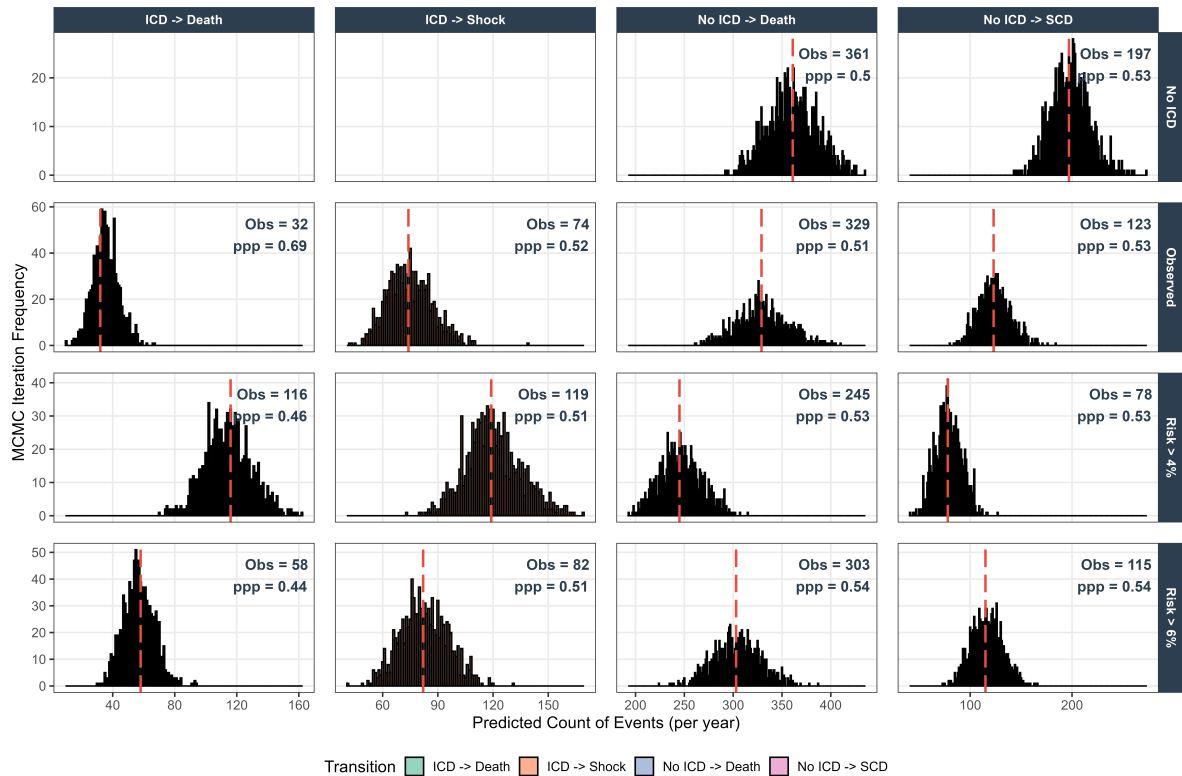
State Transition Probability Posterior Distributions



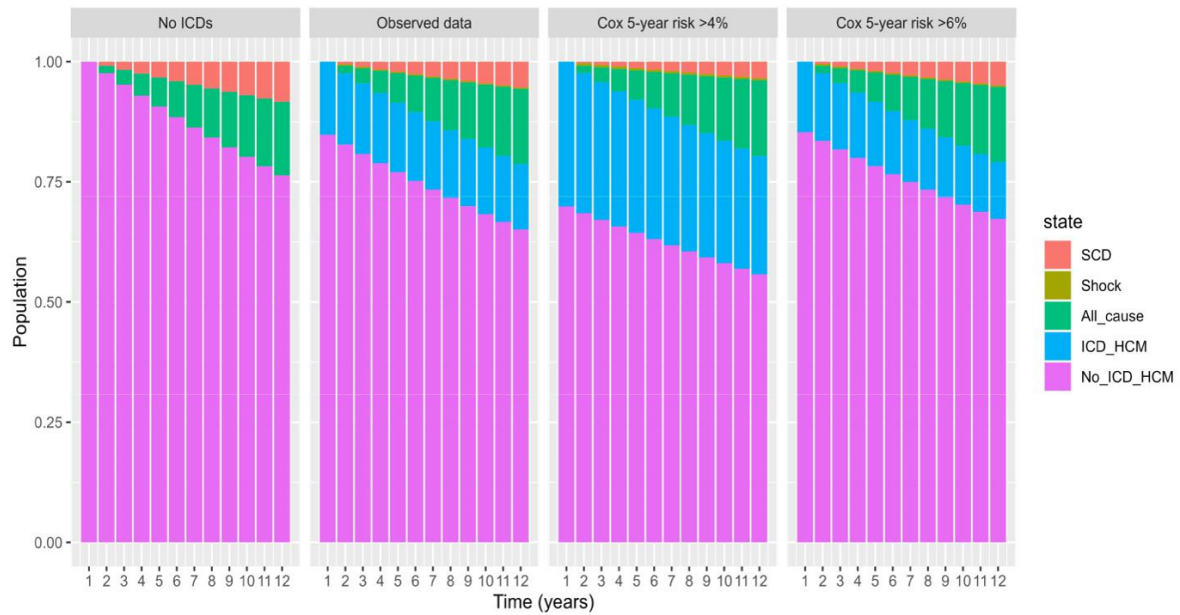
Posterior Predictive Checks

Posterior Predictive Checks: Transition Counts

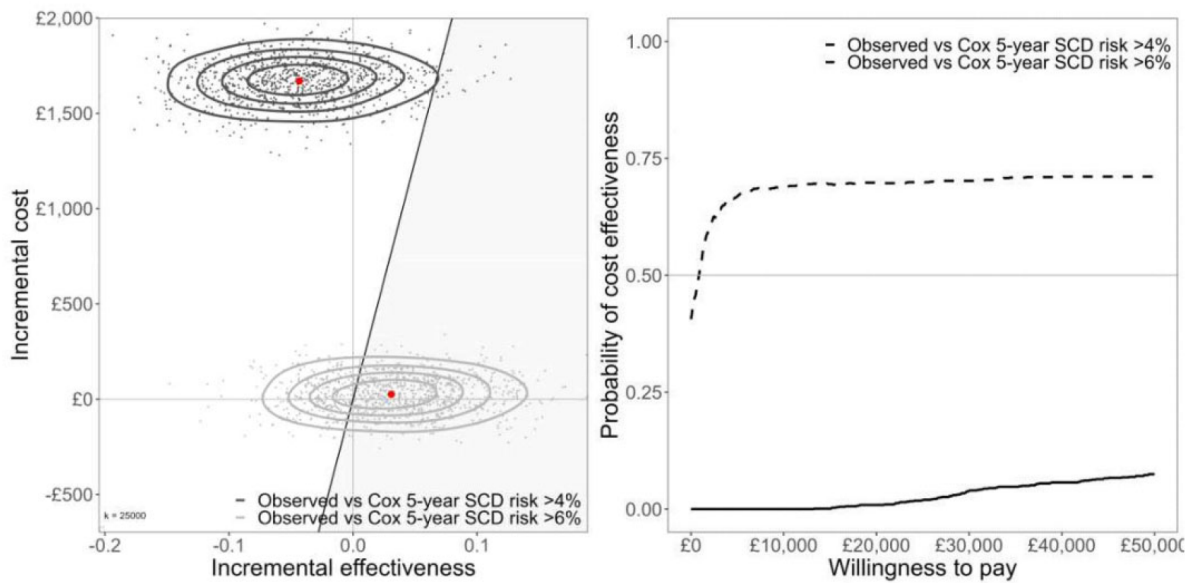
Simulated transition counts (histogram) compared to actual observed counts (red dashed line).



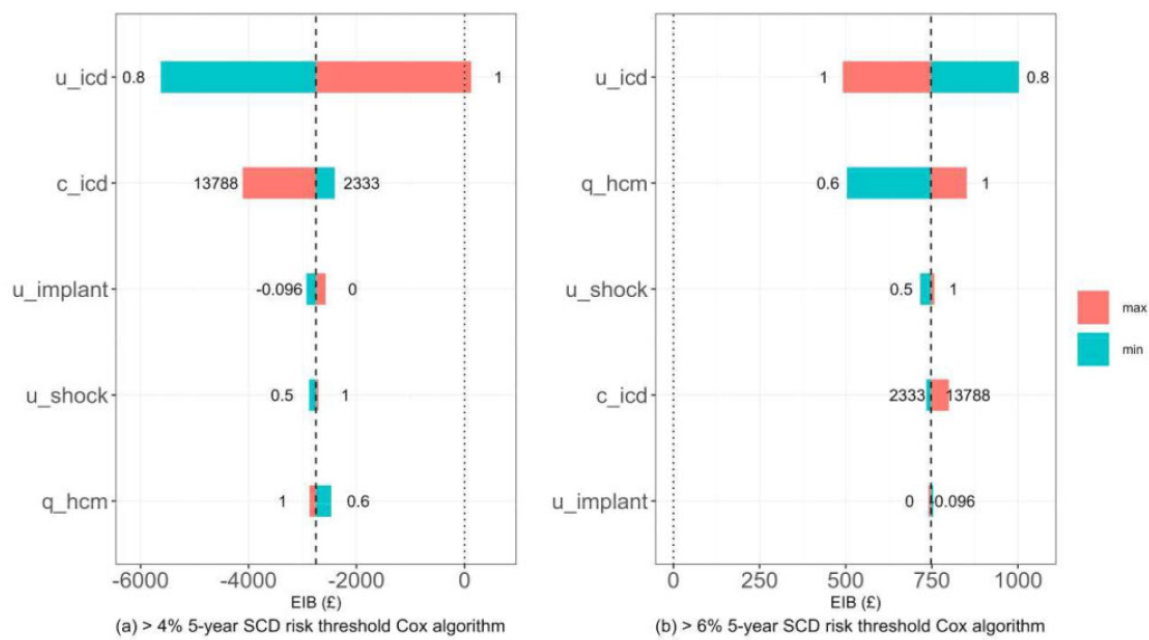
Trace Plots



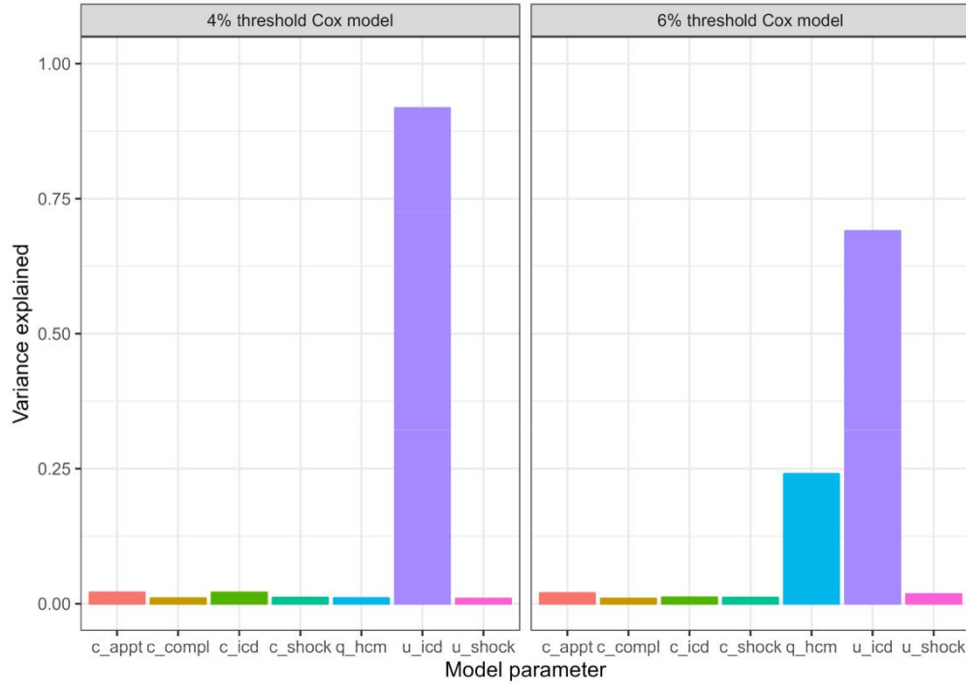
Cost-Effectiveness Analysis Plots



Deterministic Sensitivity Analysis (DSA)



Global First-Order Variance-Based Probability Sensitivity Analysis

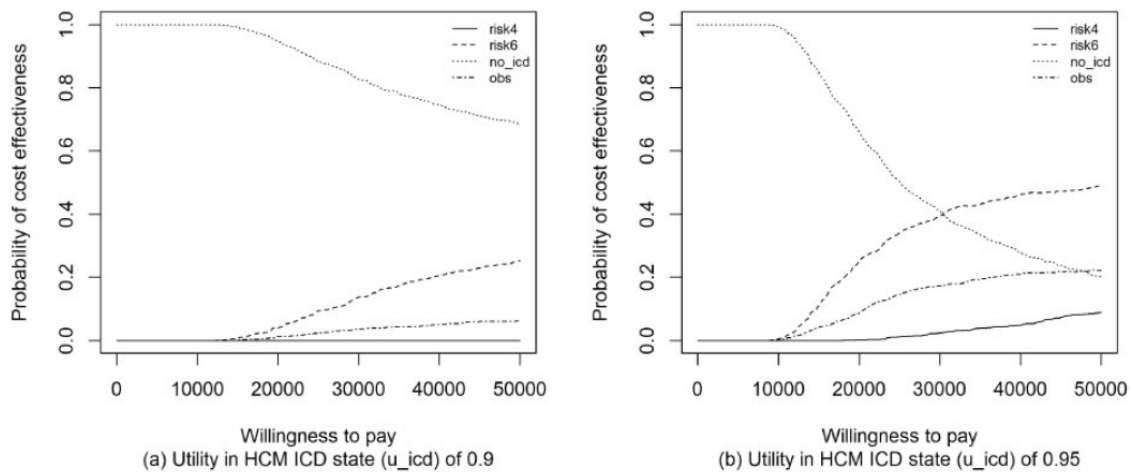


(main effect / first-order Sobol index)

$$S_i = \frac{V_{X_i}(E_{\mathbf{X}_{\sim i}}(Y | X_i))}{V(Y)}$$

- Y is the model output.
- X_i is the specific input parameter of interest.
- $\mathbf{X}_{\sim i}$ denotes all other input parameters except X_i .
- $V(Y)$ is the total unconditional variance of the model output.

Cost-Effectiveness Acceptability Curves for ICD Costs

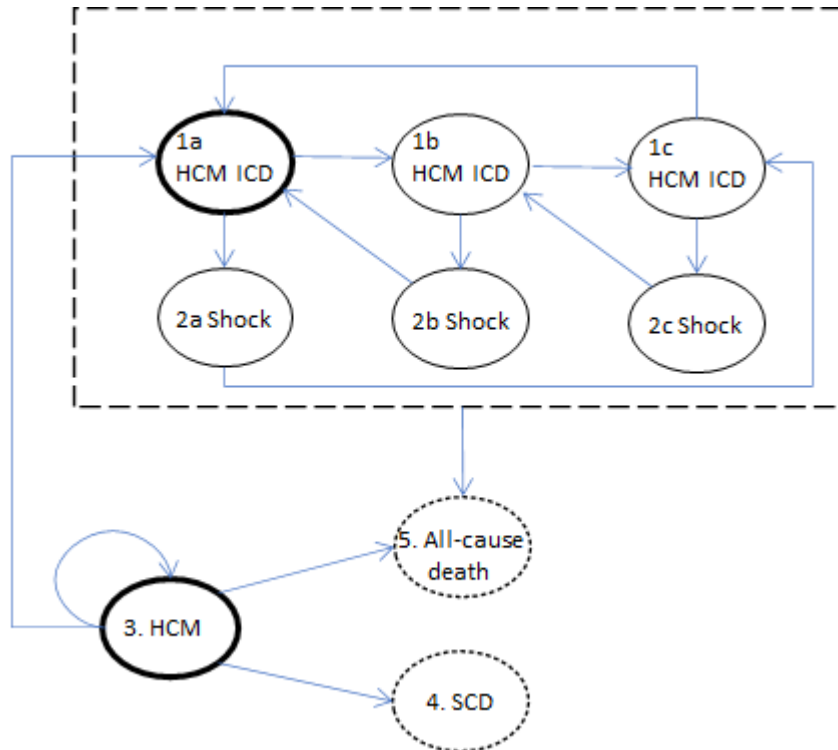


Future Work

Subcutaneous ICD (S-ICD)

- **Structural Model Extension:** Adapt the existing Markov framework to differentiate between Transvenous (TV-ICD) and Subcutaneous (S-ICD) devices.
- **Differential Cost-Effectiveness:** Evaluate the economic trade-off between the higher upfront capital costs of S-ICDs against the potential long-term savings.
- **Subgroup Stratification:** Younger cohorts?

Markov Models with Device Replacement



Expected Value of Partial Perfect Information (EVPPPI)

From Model Uncertainty to Decision Uncertainty

- **Transition to EVPPPI:** Shift the focus from global variance sensitivity to the Expected Value of Partial Perfect Information (EVPPPI).
- **Isolate Decision Drivers:** Identify which specific clinical parameters (e.g., LV wall thickness coefficients, baseline hazards) actively cause patients to cross the 6% 5-year risk threshold and alter the ICD implantation decision.
- **Quantify Research Value:** Calculate the maximum expected Net Monetary Benefit (NMB) of resolving uncertainty for these high-impact parameters to prioritize future clinical data collection.

Expected Value of Partial Perfect Information (EVPPPI)

For a specific parameter of interest (X_i) is calculated as:

! EVPPI

$$\text{EVPPI}_{X_i} = E_{X_i} \left[\max_{d \in \{0,1\}} E_{X_{\sim i}} [\text{NMB}(d, X_i, X_{\sim i})] \right] - \max_{d \in \{0,1\}} E_{\theta} [\text{NMB}(d, \theta)]$$

- X_i : The specific parameter of interest (e.g., LV wall thickness coefficient).
- $X_{\sim i}$: All other uncertain parameters in the model.
- $d \in \{0,1\}$: The binary clinical decision (Implant ICD).
- NMB: Net Monetary Benefit.

Cumulative Incidence Functions (CIFs)

! Fine-Gray Model (Subdistribution Hazards)

- $\hat{H}_{SCD}(5)$ is the baseline cumulative subdistribution hazard at 5 years.
- PI_{FG} represents the linear predictor re-fitted using a Fine-Gray penalty/weights.

$$\text{CIF}_{SCD}^{FG}(5) = 1 - \exp \left(-\hat{H}_{SCD}(5) \exp(PI_{FG}) \right)$$

! Cause-Specific Model

- Requires the integration of cause-specific hazards for SCD and all other competing events.

$$\hat{h}_{SCD}(t) \text{ and } \hat{h}_{Other}(t)$$

are the baseline hazards at time t.

$$PI_{cs,SCD} \text{ and } PI_{cs,Other}$$

are the linear predictors from their respective Cox models.

$$\text{CIF}_{SCD}^{cs}(5) = \int_0^5 \hat{h}_{SCD}(u) \exp(PI_{cs,SCD}) \exp \left(- \int_0^u [\hat{h}_{SCD}(s) \exp(PI_{cs,SCD}) + \hat{h}_{Other}(s) \exp(PI_{cs,Other})] ds \right) du$$

Conclusions

- **Individualized Risk Stratification:** The novel risk prediction model provides individualized 5-year probabilities for Sudden Cardiac Death, offering a more precise tool for clinical decision-making than previous crude guidelines.
- **Robust Economic Evaluation:** By embedding this prediction model within a Bayesian discrete-time Markov framework, we can formally quantify the cost-effectiveness of different ICD implantation thresholds under real-world uncertainty.
- **Future Work:**
 - Shifting from model uncertainty to decision uncertainty (EVPPI) will provide a more rigorous foundation for prioritizing future data collection
 - State-of-the-art device evaluation (e.g., S-ICDs).

Thanks

References

- Green, Nathan, Yang Chen, Constantinos O’Mahony, et al. 2024. “A Cost-Effectiveness Analysis of Hypertrophic Cardiomyopathy Sudden Cardiac Death Risk Algorithms for Implantable Cardioverter Defibrillator Decision-Making.” *European Heart Journal - Quality of Care and Clinical Outcomes* 10: 285–93. <https://doi.org/10.1093/ehjqcco/qcad050>.
- O’Mahony, Constantinos, Fatima Jichi, Menelaos Pavlou, et al. 2014. “A Novel Clinical Risk Prediction Model for Sudden Cardiac Death in Hypertrophic Cardiomyopathy (HCM Risk-SCD).” *European Heart Journal* 35: 2010–20. <https://doi.org/10.1093/eurheartj/eh439>.